

RESEARCH ARTICLE

A general White-Box strategy for designing thermoelectric cooling system

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Abstract

Thermoelectric cooling (TEC) is critically important in thermal management of laser modules or chips and potentially for personalized thermoregulation. The formulae for efficiency in standard textbooks can only describe the performance of a TEC module with ideal thermal conditions, that is, fixed terminal temperatures, but are unable to deal with a real TEC system where heat transfer at its interfaces with the heat source and sink are finite and with thermal resistances. Here, we define the TEC system-level performance indices, that is, the maximum cooling power, temperature difference, and coefficient of performance, by introducing a set of explicit formulae. The external heat transfer conditions are taken into account as dimensionless thermal resistance parameters. With these formulae, the TEC system performances are evaluated elegantly with errors well within $\pm 5\%$ over broad operating conditions. We further optimize the cooling power and the coefficient of performance in practical scenarios and establish a general White-Box design procedure for TEC systems, which enables a transparent design process and straightforward analysis of performance bottlenecks. A set of cooling experiments are performed to validate the analytical model and to illustrate the dependence of system design on realistic thermal conditions. By choosing the suitable TEC module parameter under given external heat transfer conditions, the cooling power can be improved by more than 100%. This work sheds some light on the integral design of TEC systems for broad applications to take full advantage of the advanced thermoelectric materials in the cooling field.

KEYWORDS

system-level performance, thermoelectric cooling, White-Box design

1 | INTRODUCTION

Thermoelectric coolers (TECs) can realize heat pump capabilities over finite negative temperature differences at the cost of electric power, with faster response and smaller size compared with conventional cooling systems. Their applications are now not limited to localized cooling for hot spots in laser and electronic cooling¹ and personalized thermoregulation,^{2,3} but also involving large space heating and cooling.^{4–6} Three performance indices, that is, the maximum temperature difference ($\Delta T_{\max}$), the maximum cooling power ($Q_{c,\max}$), and the maximum coefficient of performance ($COP_{\max}$), are used to calibrate the cooling capability of a TEC device or system. In the textbooks and handbooks, the cooling performance indices of a TEC device are defined under given hot side (T_h) and cold side (T_c) temperatures as⁷

$$\begin{cases} Q_{c,\max} = 2N \left(\frac{S^2 T_c^2}{2R} - K \Delta T \right) = 2N \frac{Z T_c^2 / 2 - \Delta T}{R_{t,l}}, & (A) \\ \Delta T_{\max} = \frac{Z T_c^2}{2} = T_h - \frac{\sqrt{1 + 2Z T_h} - 1}{Z}, & (B) \\ COP_{\max} = \frac{T_c}{\Delta T} \frac{\sqrt{1 + Z \bar{T}} - \frac{T_h}{T_c}}{\sqrt{1 + Z \bar{T}} + 1}, & (C) \end{cases} \quad (1)$$

where ΔT and $\bar{T}$ are the temperature difference and the average temperature between T_h and T_c , respectively. $Z = \frac{S^2 \sigma}{\kappa}$ is the figure of merit of the thermoelectric material, with S , σ , and κ denoting the Seebeck coefficient, the electrical conductivity, and the thermal conductivity, respectively. R and $R_{t,l}$ are respectively the electrical and thermal resistance of the TEC leg calculated by $R = \frac{L}{\sigma A}$ and $R_{t,l} = \frac{L}{\kappa A}$, and L , A , and N are the leg length, cross-sectional area, and the number of thermocouples. K is the thermal conductance, which is the reciprocal of $R_{t,l}$. The material properties are assumed to be temperature independent to derive Equation (1).

In a practical implementation of a thermoelectric cooling (TEC) system, however, the terminal temperatures of the TEC module are not fixed, but usually determined by the coupling with the heat source or sink. As a result of the nonzero thermal resistance of the heat exchangers and the induced finite heat transfer rates, the terminal temperatures of the TEC module are different from those of the heat source and sink, as illustrated in Figure 1A. To establish a specified temperature difference (ΔT) between the heat source and the heat sink, a larger temperature difference ($\Delta T'$) within the TEC module is needed (Figure 1B), leading to a degradation of the system level performance compared to the device level performance defined by Equation (1).

The system-level studies on TECs could be generally classified into three topics. The first topic is to calculate the cooling performances of TECs under real working conditions, with known module dimensions and material properties^{8–12} or module performance data under standard test conditions.^{13–16} An analytical-numerical combined method is often used. First, a heat balance analysis is performed at the thermal interfaces, to derive primary expressions for the cooling power and input electric power. Then a numerical scanning approach is used to seek the maximum cooling power,^{17–20} temperature difference,^{21,22} or COP with respect to the electric current. Only a few analytical formulations on the key performance indices are deduced under very limited conditions^{23–26} due to the high complexity. Based on the first one, the second topic is optimizing the design of TEC systems in real thermal circumstances. The aim is to find the design parameters for the maximum cooling power,^{27–31} COP,^{28–31} or locally maximum COP under given cooling power.^{23,32,33} The key is to achieve the optimum thermal resistance allocation over the three parts in the thermal network, that is, the TEC module, heat source, and heat sink. Many studies have reported that the hot side thermal resistance plays a more significant role than that of the cold side one on the system-level cooling performance.^{20,23,27,34–37} In the above

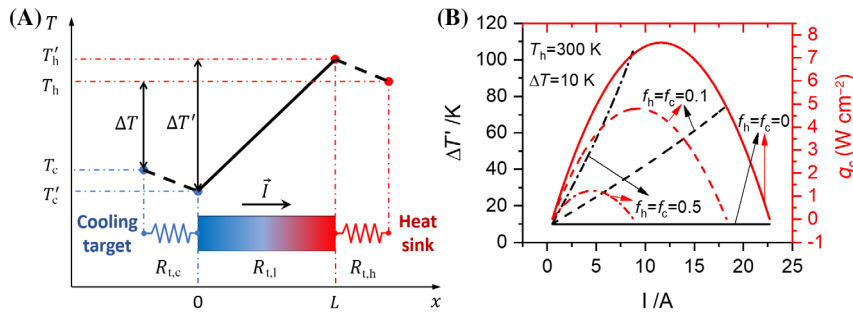


FIGURE 1 (A) The thermal network and qualitative temperature distribution in a TEC system. (B) Evolutions of $\Delta T'$ and cooling power density with the electric current under different external thermal resistance conditions (f_h and f_c denote the thermal resistance ratios at the hot and cold sides, respectively)

two topics, the thermal resistances or heat transfer coefficients at the hot and cold sides of the TEC module are denoted as abstract symbols in the analyses and was never inspected in detail. Finally, the third topic concerns the technical advances of heat exchangers,^{38–49} as well as the minimization of the cost of real TEC systems.^{50,51}

Despite the wide range of applications of TEC, a general explicit formula for TEC system performance is yet to be found. Although numerical optimization with brutal force calculations does work, it fails to provide physical insights on how intercoupled properties can be tuned for improved performances. Indeed there is a strong demand for a set of general formulations on a real TEC system's performance indices to offer a straightforward way to uncover the whole picture and guide the system design, just like what we have done for TEG systems.⁵² Therefore, in this work our concern is on evaluating the performance of TEC systems based on state-of-the-art materials. We will not look into the development of thermoelectric materials, which have already gained extensive attentions.^{53,54} Our discussions will be organized in progressive order as follows. First, we will construct an analytical model to derive the standard formulae of the performance indexes of TEC systems. Based on the analytical model, we will look into three critical issues in TEC system design and operation one by one. So second, we will focus on maximizing the cooling power under given heat transfer conditions. Third, we will realize the optimization of COP under given heat transfer conditions

2 | ANALYTICAL MODEL

In this section, we will deduce the formulae of the performance indexes of a TEC system, starting from the governing equation and heat balance conditions at the thermal interfaces,

$$\begin{cases} \kappa \frac{d^2 T}{dx^2} + \frac{I^2}{\sigma A^2} = 0 & \text{(A)} \\ -\kappa A \frac{dT}{dx} + ST'_c I = \frac{T_c - T'_c}{R_{t,c}}, x=0 & \text{(B)} \\ -\kappa A \frac{dT}{dx} + ST'_h I = \frac{T'_h - T_h}{R_{t,h}}, x=L & \text{(C)} \end{cases} \quad (2)$$

where the relevant parameters are illustrated in Figure 1. For the sake of concision, only one TEC leg is considered in the analysis. It can be easily extended to a module with multiple legs. The cooling target temperature and the external thermal resistance at the cold side are denoted as T_c and $R_{t,c}$, while those at the heat sink side as T_h and $R_{t,h}$, respectively. The terminal temperatures of the leg are represented by T'_c and T'_h . I is the input electric current, and other parameters are the same as those in Equation (1). The material properties are assumed to be independent of temperature.

First, the temperature profile along the current direction is derived from Equation (2),

$$\begin{cases} T = -\frac{0.5I^2 R R_{t,l}}{L^2} x^2 + C_1 x + C_2 \\ C_1 = \frac{1}{L} \frac{c_1 f_h I^2 R R_{t,l} + 0.5c_1 c_2 I^2 R R_{t,l} + S R_{t,l} I (f_h T_c + f_c T_h) + \Delta T}{f_h + f_c + c_1 c_2} \\ C_2 = \frac{1}{c_1} \left\{ \frac{f_c [c_1 f_h I^2 R R_{t,l} + 0.5c_1 c_2 I^2 R R_{t,l} + S R_{t,l} I (f_h T_c + f_c T_h) + \Delta T]}{f_h + f_c + c_1 c_2} + T_c \right\} \\ c_1 = 1 + f_c S R_{t,l} I \\ c_2 = 1 - f_h S R_{t,l} I \end{cases} \quad (3)$$

and cooling power, using the peak cooling power as a base point. Then, we will discuss the optimal thermal resistance allocation, which is critical for the COP optimization under a given cooling power. A general White-Box design procedure is finally established for TEC systems based on the former topics, and a summary is made at last.

where the external thermal resistances are represented by the dimensionless thermal resistance parameters as $f_h = R_{t,h}/R_{t,l}$ and $f_c = R_{t,c}/R_{t,l}$. Here, $R_{t,l}$ is the thermal resistance of the TEC leg. Based on Equation (3), the terminal temperatures of the leg and the temperature difference within it are directly given by, By defining a

$$\begin{cases} T'_c = C_2 \\ T'_h = -0.5I^2RR_{t,l} + C_1L + C_2 \\ \Delta T' = \frac{\Delta T + SR_{t,l}I(f_hT_c + f_cT_h + f_hf_cI^2RR_{t,l}) + 0.5I^2RR_{t,l}(f_h - f_c)}{f_h + f_c + c_1c_2} \end{cases} \quad (4)$$

dimensionless current $I^* = SR_{t,l}I$, we obtain the expressions for the cooling power, the input electric power, and the coefficient of performance, respectively,

$$Q_c = \frac{T_c - T'_c}{R_{t,c}} = \frac{1}{ZR_{t,l}} \frac{0.5f_hI^{*3} - 0.5[1 + 2f_h(1 + ZT_c)]I^{*2} + ZT_cI^* - Z\Delta T}{f_h + f_c + (1 + f_cI^*)(1 - f_hI^*)} \quad (5)$$

$$P = S\Delta T'I + I^2R = \frac{I^*}{ZR_{t,l}} \frac{-0.5(f_h - f_c)I^{*2} + [1 + f_h + f_c + Z(f_hT_c + f_cT_h)]I^* + Z\Delta T}{f_h + f_c + (1 + f_cI^*)(1 - f_hI^*)} \quad (6)$$

$$COP = \frac{Q_c}{P} = \frac{1}{I^*} \frac{0.5f_hI^{*3} - 0.5[1 + 2f_h(1 + ZT_c)]I^{*2} + ZT_cI^* - Z\Delta T}{-0.5(f_h - f_c)I^{*2} + [1 + f_h + f_c + Z(f_hT_c + f_cT_h)]I^* + Z\Delta T} \quad (7)$$

Until now, there is no essential difference between our derivation and those in the previous studies,^{8–22} except the adoption of different dimensionless parameters. However, it is worth pointing out that a concise appearance of the expression could make a difference. It is possible to push forward the derivation and deduce approximate analytical formulae of the performance indices. We will depart from the standard way of numerically solving the above equations and move on with analytical derivations.

2.1 | Maximum cooling power

We will first look into the cooling power expression. Given material properties, module dimensions, and thermal conditions, the cooling power is solely dependent on the dimensionless current, I^* . Taking $dQ_c/dI^* = 0$ will give the optimum dimensionless current for maximum cooling power. Although an exact formula of the optimum I^* is unavailable due to the complicated form of the cooling power expression, we will approximately

construct the expected formula by ignoring some minor terms in Equation (5). As a preliminary attempt, we obtain a simple form of the optimum I^* by ignoring the cubic term in the numerator and the whole denominator in the derivative of Q_c with respect to I^* as follows,

$$I_{q_{c,max},1}^* = \frac{ZT_c}{1 + 2f_h(1 + ZT_c)} \quad (8)$$

Substituting Equation (8) into Equation (5), the maximum cooling power is calculated by

$$Q_{c,max} = \frac{0.5\alpha_1 ZT_c^2 - \alpha_2 \Delta T}{\alpha_3 R_{t,l}} \quad (9)$$

where we introduce three factors, that is,

$$\alpha_1 = 1 + \frac{f_h ZT_c}{[1 + 2f_h(1 + ZT_c)]^2}$$

$$\alpha_2 = 1 + 2f_h(1 + ZT_c)$$

$$\alpha_3 = (1 + 2f_h)[1 + (f_h + f_c)(1 + ZT_c)]$$

Equation (9) is indeed similar in form to Equation (1A), considering practical heat transfer conditions by including the dimensionless thermal resistance parameters in the above factors. We can rewrite Equation (9) in its intensive forms, which are more convenient to use in further analyses,

$$\begin{cases} q_{cl,max} = \frac{1}{A} \frac{0.5\alpha_1 ZT_c^2 - \alpha_2 \Delta T}{\alpha_3 R_{t,l}}, & (A) \\ q_{c,max} = \frac{FF}{A} \frac{0.5\alpha_1 ZT_c^2 - \alpha_2 \Delta T}{\alpha_3 R_{t,l}}, & (B) \end{cases} \quad (10)$$

where Equation 10A is the maximum cooling power density based on the leg cross-sectional area, while Equation 10B is that based on the substrate area of a module accounting for the fill factor (FF) of the legs. To check the accuracy of the approximate formulae of the

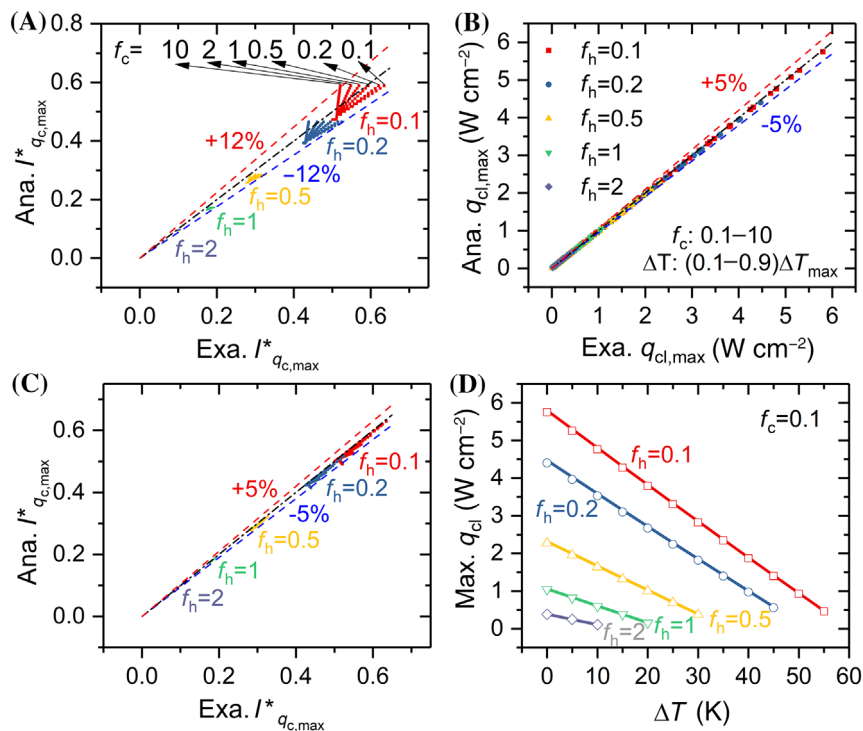


FIGURE 2 Comparisons of the analytical results of the dimensionless electric current by (A) Equation (8), (C) Equation (11) for maximum cooling power density (B) by Equation (10A) with numerical data. (D) Evolutions of the maximum cooling power density with ΔT

optimum I^* and the maximum cooling power density, a direct comparison between the results calculated by Equations (8) and (10A) and those by numerical solutions are illustrated in Figure 2A,B. In a previous study,⁵² we have estimated the external thermal resistance parameters f_h and f_c to be generally within 0.05–10 in realistic thermal circumstances for thermoelectric power generator. Therefore, we will follow the above range when performing formula validations in this study. All the calculations in this work are based on the thermoelectric properties of conventional materials Bi₂Te₃, that is, $S = 200 \mu V K^{-1}$, $\sigma = 1e5 S m^{-1}$, $\kappa = 1.5 W m^{-1} K^{-1}$ and leg dimensions of $L = 2 mm$ and $A = 4 mm^2$, unless otherwise noted. As a rather rough approximation, Equation (8) leads to moderate evaluations of the optimum I^* with errors within $\pm 12\%$, while Equation (10A) leads to much better evaluations of the maximum cooling power density with errors within 5%. This does make sense because, in the vicinity of the exact maximum cooling power, the cooling power is not sensitive to I^* due to the nearly zero derivative. Nevertheless, it is essential to evaluate the optimum I^* with a higher accuracy for other reasons, which will be discussed later. Here, we have come up with a better optimum I^* formula over Equation 8 by multiplying a correction factor ξ to it, as a result of a partial retrieve of the effects of the cubic term and the

denominator in Equation 5 (see Supporting Information (SI): The electric current formula for maximum cooling power for a detailed derivation),

$$I^*_{q_{c,max},2} = \xi I^*_{q_{c,max},1} \quad (11)$$

where

$$\xi = \frac{1 + \zeta \frac{\Delta T}{T_c}}{1 + \frac{1}{2} \left(\zeta - 3 \frac{f_h}{\alpha_2} \right) I^*_{q_{c,max},1}}$$

and

$$\zeta = \frac{f_c - f_h}{1 + f_h + f_c}$$

Equation (11) demonstrates a noticeable improvement in the evaluation of the optimum I^* for maximum cooling power, with errors within $\pm 5\%$ for the same operation range (see Figure 2C). While it would not give rise to a visible improvement in evaluating the maximum cooling power since Equation (9) is already very accurate under most conditions. Consequently, Equation (9) will be the final version of the maximum cooling power in

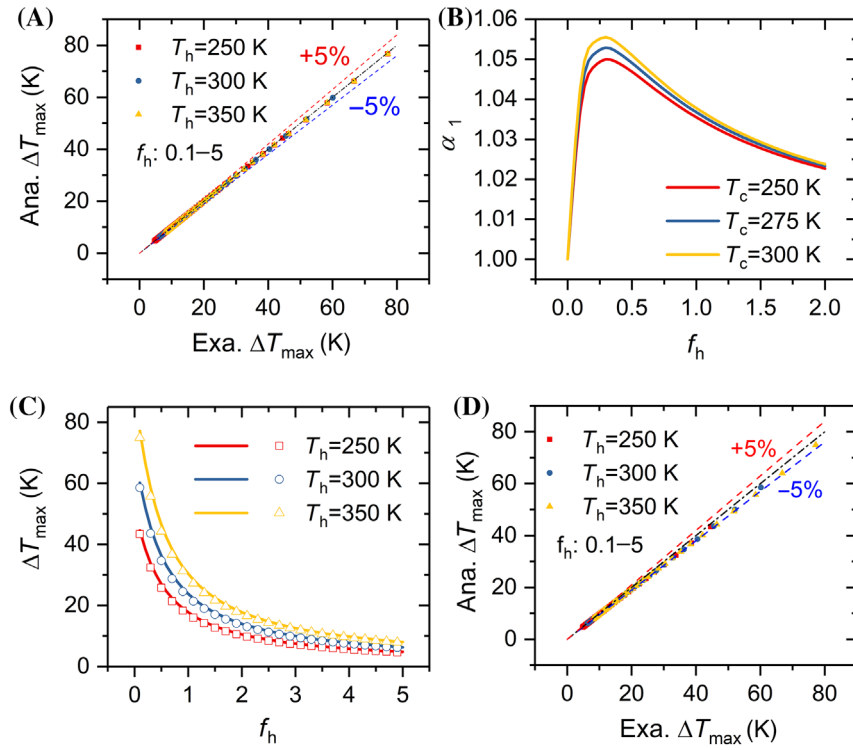


FIGURE 3 (A) Comparison of analytical results of the maximum temperature difference by Equation (12) with numerical data. (B) Evolutions of α_1 with f_h under different cold side temperatures. (C) Evolutions of the maximum temperature difference with f_h . (D) Comparison of analytical results of the maximum temperature difference by Equation (13) with numerical data

this work. Figure 2D presents the evolution of the maximum cooling power density with the temperature difference under a given f_c and a series of f_h . From Figure 2C,D it is clearly seen that the optimum I^* , the maximum cooling power and the achievable temperature difference are all significantly reduced by f_h .

2.2 | Maximum temperature difference

Now we will deduce an approximate formula of the maximum temperature difference based on Equation (9). Letting $Q_{c,\max} = 0$, the maximum temperature difference is obtained as

$$\Delta T_{\max} = \frac{\alpha_1 Z T_c^2}{2\alpha_2} \quad (12)$$

Equation (12) is similar in form with Equation (1B), that is, $\Delta T_{\max} = \frac{Z T_c^2}{2}$ for a TEC module under an ideal thermal condition (fixed temperature at the two terminals), and it can estimate the maximum temperature difference with very high accuracy under practical thermal conditions (Figure 3A). Nonetheless, it is an implicit form because the coefficients α_1 and α_2 themselves depend on

the cold side temperature. To transform it into an explicit form for a straightforward using purpose, we will have to further approximate α_1 as 1 to give

$$\begin{aligned} \Delta T_{\max} &= \frac{Z T_c^2}{2[1 + 2f_h(1 + ZT_c)]} \\ &= \beta_1 \left(T_h - \frac{\sqrt{1 + 2ZT_h + \beta_2^2 - 1}}{Z} \right) \end{aligned} \quad (13)$$

where $\beta_1 = \frac{1+2f_h}{1+4f_h}$, and $\beta_2 = \frac{2f_h Z T_h}{1+2f_h}$. In fact α_1 is a parameter very close to 1 in common situations as seen in Figure 3B, so the above approximation will only lead to a minor error in the evaluation of the maximum temperature difference. The explicit form in Equation (13) is also close to the device level formula, which would be helpful for its acceptance by the thermoelectric community. Figure 3C,D give a comparison between the maximum temperature differences calculated by Equation (13) and those by numerical solutions in a f_h range of 0.1–5 under three different heat sink temperatures. As expected, a satisfying agreement is achieved with errors within 5%. The dimensionless current for maximum temperature difference is also calculated using Equations (8) and (11).

2.3 | Maximum COP

Now we have come to the last performance index, that is, the maximum COP. After some cumbersome algebra by letting $d(COP)/dI^* = 0$, we arrive at a formula for the maximum COP

$$COP_{\max} = \frac{T_c}{\Delta T} \frac{\sqrt{1 + Z\hat{T}^*} - \frac{T_h}{T_c} + \gamma_1}{\sqrt{1 + Z\hat{T}^*} + 1} + \gamma_2 \quad (14)$$

and the corresponding dimensionless current

$$I_{COP_{\max}}^* = \frac{\Delta T}{\hat{T}} \left(\sqrt{1 + Z\hat{T}^*} + 1 \right) \quad (15)$$

where

$$\hat{T}^* = \frac{\hat{T}}{1 + F_h + F_c}$$

$$\hat{T} = \bar{T} + 0.5\Delta T^*$$

$$\Delta \hat{T}^* = (F_h - F_c)\Delta T^*$$

$$\Delta T^* = \frac{\Delta T}{1 + F_h + F_c}$$

$$F_h = f_h(1 + ZT_c)$$

$$F_c = f_c(1 + ZT_h)$$

$$\gamma_1 = -\frac{\Delta \hat{T}^*}{T_c}$$

$$\gamma_2 = f_h \frac{\Delta T^*}{\hat{T}}$$

Again, Equation (14) is similar in form to Equation (1C) and offers a straightforward evaluation of the maximum COP of a real TEC system with known dimensionless thermal resistance parameters. Figure 4 illustrates the accuracies of Equations (14) and (15) in evaluating the maximum COP as well as the corresponding dimensionless current (Figure 4A,B), and the evolutions of them with respect to the temperature difference under different f_h and f_c (Figure 4C,D). Excellent agreements are achieved between analytical and numerical results. As expected, the maximum COP decreases with both f_h , f_c , and ΔT , however, the corresponding dimensionless current seems to be linear with ΔT but independent of f_h and f_c .

As a summary of this section, the explicit formulae of the three performance indexes of a TEC system are written together here,

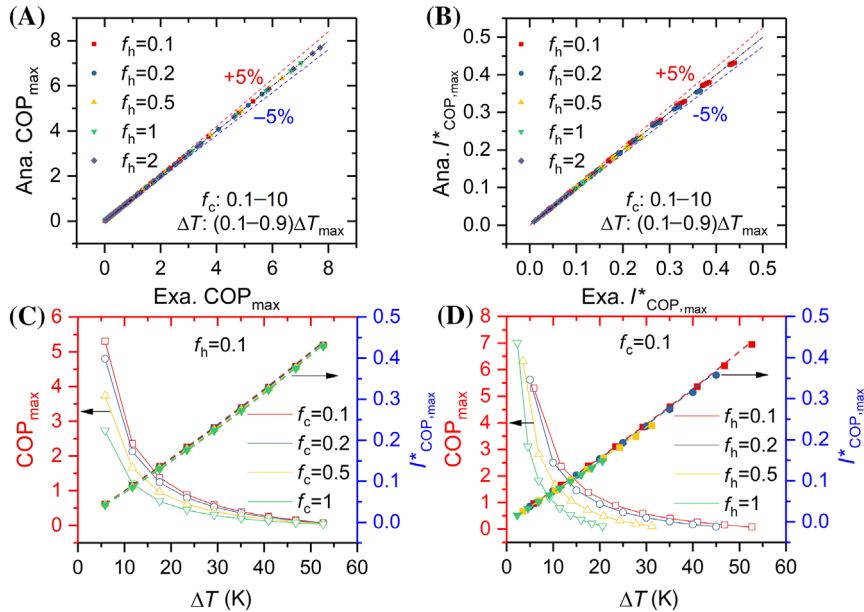


FIGURE 4 Comparisons of analytical results of (A) the maximum COP by Equation (14) and (B) the corresponding dimensionless electric current by Equation (15) with numerical data; evolutions of the maximum COP and the corresponding dimensionless electric current with ΔT under (C) constant f_h and (D) constant f_c

$$\begin{cases} Q_{c,\max} = \frac{0.5\alpha_1 Z T_c^2 - \alpha_2 \Delta T}{\alpha_3 R_{t,l}}, & (A) \\ \Delta T_{\max} = \beta_1 \left(T_h - \frac{\sqrt{1 + 2Z T_h + \beta_2^2} - 1}{Z} \right), & (B) \\ COP_{\max} = \frac{T_c}{\Delta T} \frac{\sqrt{1 + Z \hat{T}^*} - \frac{T_h}{T_c} + \gamma_1}{\sqrt{1 + Z \hat{T}^*} + 1} + \gamma_2, & (C) \end{cases} \quad (16)$$

and the corresponding dimensionless current conditions,

$$\begin{cases} I_{q_{c,\max}}^* = I_{\Delta T_{\max}}^* = \frac{\xi Z T_c}{\alpha_2}, & (A) \\ I_{COP_{\max}}^* = \frac{\Delta T}{\hat{T}} \left(\sqrt{1 + Z \hat{T}^*} + 1 \right), & (B) \end{cases} \quad (17)$$

or in their dimensional forms

$$\begin{cases} I_{q_{c,\max}} = I_{\Delta T_{\max}} = \frac{\xi S T_c}{\alpha_2 R}, & (A) \\ I_{COP_{\max}} = \frac{S \Delta T^*}{R} \frac{1}{\sqrt{1 + Z \hat{T}^*} - 1}, & (B) \end{cases} \quad (18)$$

Under ideal thermal conditions with zero thermal resistance at both ends, by substituting $f_h = f_c = 0$ into the above formulae, Equations (16) and (18) go back to the textbook formulae, implying a good consistency between the device and system-level analyses.

The above analysis is based on the assumption that the properties of the material are temperature independent, which is not true for real thermoelectric materials. However, compared to thermoelectric power generators, TEC systems suffer from much narrower temperature differences, which are generally on the order of tens of degrees. Consequently, the temperature dependences of material properties are expected to have a minor effect on the system performance compared to the external heat transfer conditions, hence, the average properties could be adopted in the analytical calculations. (See SI: Temperature dependence of thermal/electric properties for detailed information). Moreover, remember that the analysis is performed on a single leg, while a real TEC module is made up of multiple p-type and n-type legs. We have demonstrated that by introducing some kind of average techniques, we can easily give accurate performance evaluations of real TEC couples based on the above analytical framework. (See SI: Performance evaluation of a TE couple for detailed information).

3 | RESULTS AND DISCUSSION

3.1 | Maximization of cooling power under given h_h , h_c , and ΔT

In the above section, we have discussed the maximization of the performance indexes of a TEC system with respect to the input electric current, by keeping all other parameters constant. They are useful in the performance evaluation and electric current optimization of an existing system, but cannot be directly used for the design of a projected system with more design freedom. In the design of a TEC system, the heat exchangers at the hot and cold sides as well as their auxiliaries would be governed not only by the performance requirements, but also by other practical constraints, including space and cost, and so on. Therefore, when designing TEC, the proper question to ask is that, once the heat transfer conditions are given, what is the upper limit of the cooling power density under a specific temperature difference, and how to determine the optimal module parameters to achieve it? In this section, we will focus on answering this question with an analytical approach.

Usually the heat transfer conditions can be written as thermal resistances, or the heat transfer coefficients based on the substrate area A_s of the module denoted as h_h and h_c , where the thermal resistances at the hot and cold sides could be written as $R_{t,h} = 1/h_h A_s$ and $R_{t,c} = 1/h_c A_s$. The thermal resistance of the module is $R_{t,m} = L/2N\kappa A$, so the dimensionless thermal resistance parameters could be expressed in terms of h_h , h_c , and the module parameters as

$$f_h = \frac{R_{t,h}}{R_{t,m}} = \frac{2N\kappa A}{h_h A_s L} = \frac{\kappa}{h_h L^*} \quad (19)$$

$$f_c = \frac{R_{t,c}}{R_{t,m}} = \frac{\kappa}{h_c L^*} \quad (20)$$

where the undetermined module parameters (the leg length and the fill factor) are lumped as $L^* = \frac{L}{FF}$, which is already used and named as the engineering leg length in our previous work⁵² and also used as a single design parameter in Reference 22. Substituting Equations (19) and (20) into Equation (10B), we obtain the underlying relationship between the maximum cooling power density and L^* ,

$$q_{c,\max} = \frac{\kappa}{L^*} \frac{0.5\alpha_1(L^*) Z T_c^2 - \alpha_2(L^*) \Delta T}{\alpha_3(L^*)} \quad (21)$$

where $\alpha_1(L^*)$, $\alpha_2(L^*)$, and $\alpha_3(L^*)$ are all functions of L^* , so $q_{c,\max}$ is also a single-parameter function of L^* . By letting $dq_{c,\max}/dL^* = 0$, and approximating $\alpha_1(L^*)$ to 1 again,

as is done in the derivation of the explicit formula of $\Delta T_{\max}$, the optimal L^* for the peak cooling power density under given h_h and h_c is given by

$$L_{q_{c,\text{peak}}}^* = \frac{\kappa}{h_h} \phi \quad (22)$$

where

$$\phi = \frac{1 + ZT_c}{\varphi} \left\{ \frac{2\Delta T}{T_c} + \sqrt{\frac{ZT_h}{1 + ZT_c} \left[\varphi \left(1 + \frac{h_h}{h_c} \right) + \frac{2\Delta T}{T_c} \right]} \right\}$$

and

$$\varphi = 0.5ZT_c - \frac{\Delta T}{T_c}$$

The estimations of the optimal L^* and the peak cooling power density by Equations (21) and (22) are validated over a broad range of the heat transfer coefficients from 5 to 5000 W m⁻² K⁻¹, which is able to cover various scenarios, including natural air convection, forced convection, and even liquid–vapor phase change heat transfer. As indicated in Figure 5A,B, the optimal L^* is in general evaluated within $\pm 15\%$, while the peak cooling power density calculated by substituting Equation (22)

into Equation (21) shows negligible error all over the whole heat transfer and temperature difference range. The fact that a rough estimation of the optimal L^* still leads to an excellent estimation of the peak cooling power density is illustrated in detail in SI (Dependence of cooling power density on L^*). Equation (22) suggests that the optimal L^* decreases with h_h but increases with the ratio h_h/h_c , so it could rise to tens of centimeters under very low h_h and h_c (see Figure 5A). As a reference, we present in Figure 5C the scope of L^* for the commercially available TEC modules,⁵⁵ as well as the relationship between the maximum cooling power density ($\Delta T = 0$) and L^* over more than 370 types of products. It is noted that the values of L^* in most off-the-shelf TEC modules are of the order of magnitude of 0.1 cm and the maximum one is close to 2 cm. Although a smaller L^* is preferred for a higher cooling power density at the device level (see Figure 5C), we clearly see that modules with small L^* s, or even all of the currently off-the-shelf modules may not be suitable for cooling use in weak heat transfer situations. In fact, this shows the importance of the heat exchanger on the thermal performance of TEC systems, as confirmed by a recent case study on wearable TECs,³ in which a higher cooling performance was achieved by using a much lower density of TE legs, that is, a much higher L^* . Apparently, Equation (22) can be adopted as a guideline for module manufacturers as well

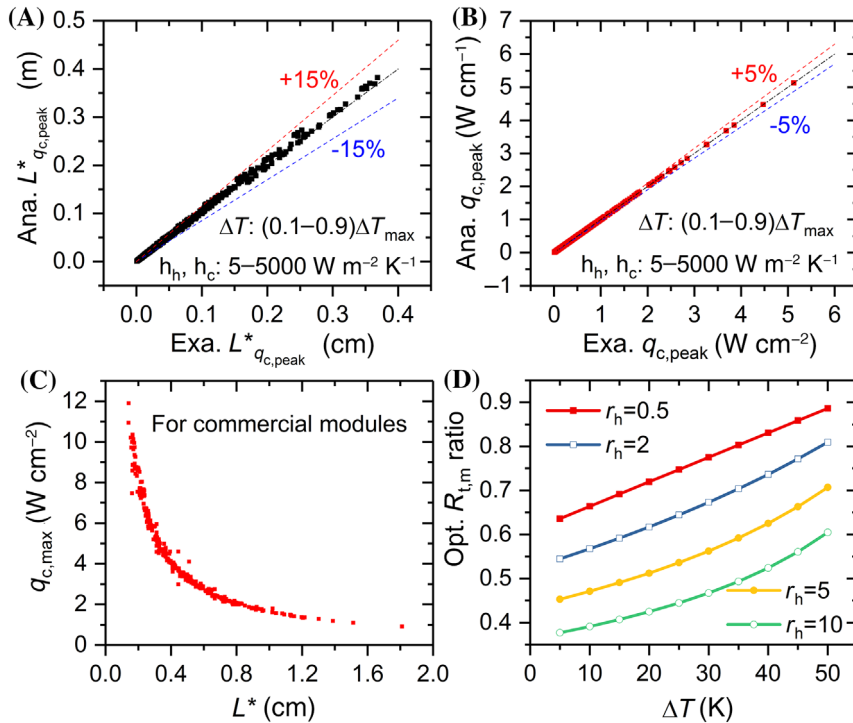


FIGURE 5 Validations of the analytical formulae of (A) the optimal L^* for (B) peak cooling power density under given h_h and h_c by numerical data; (C) available L^* range and the nominal maximum cooling power density for currently commercial modules; (D) evolutions of the optimal thermal resistance ratio of the module for peak cooling power density with ΔT and r_h

as for TEC system designers to achieve higher cooling power densities under limited heat transfer conditions.

Combining Equations (19), (20), and (22), we deduce an explicit formulation on the optimal thermal resistance ratio of the module to the total thermal network for maximum cooling power density,

$$r_{R_{t,m},q_{c,peak}} = \frac{1}{1 + (f_h + f_c)_{q_{c,peak}}} = \frac{\phi}{1 + r_h + \phi} \quad (23)$$

where $r_h = h_h/h_c$. Once the material properties, the heat sink temperature (T_h), the temperature difference (ΔT) and the heat transfer allocation ratio (r_h) are specified, we can directly determine the optimal thermal resistance ratio for maximum cooling power using Equation (23). Figure 5D shows the evolutions of the optimal thermal resistance ratio with ΔT under different r_h conditions. It indicates that the optimal thermal resistance ratio increases with ΔT and decreases with r_h . Equation (23) leads to guidelines of thermal resistance match in agreement with but more general than the numerically induced ones by Reference 27.

3.2 | Maximization of COP under given h_h , h_c , q_c , and ΔT

In this section, the COP is the emphasis in the design of TEC systems in addition to the basic requirements (Q_c

and ΔT), since it really matters in the long-term running of such systems. We will start with a close look on the COP at the maximum cooling power. Figure 6A presents typical evolutions of COP and cooling power density with the dimensionless electric current under specified operating conditions. It implies that the maximum COP point is accompanied by a relatively limited cooling power density. ^{15,56} On the other hand, the maximum cooling power suffers from a very low COP.

To estimate the COP at the maximum cooling power, we derive a formula by substituting Equation (8) into Equation (7),

$$COP_{q_{c,max}} = \frac{0.5 - \frac{1}{I_{q_{c,max}}^*} \frac{\Delta T}{T_c}}{\chi + \frac{\Delta T}{T_c}} \quad (24)$$

where

$$\chi = \frac{1 + F_h + F_c}{1 + 2F_h}$$

Equation (24) evaluates the aimed COP with errors within $\pm 20\%$ over a broad operating conditions (see Figure 6B). Figure 6C presents evolutions of the ratio of the COP at the maximum cooling power to the maximum COP with the dimensionless ΔT , suggesting a low COP ratio especially under a relatively low ΔT , regardless of

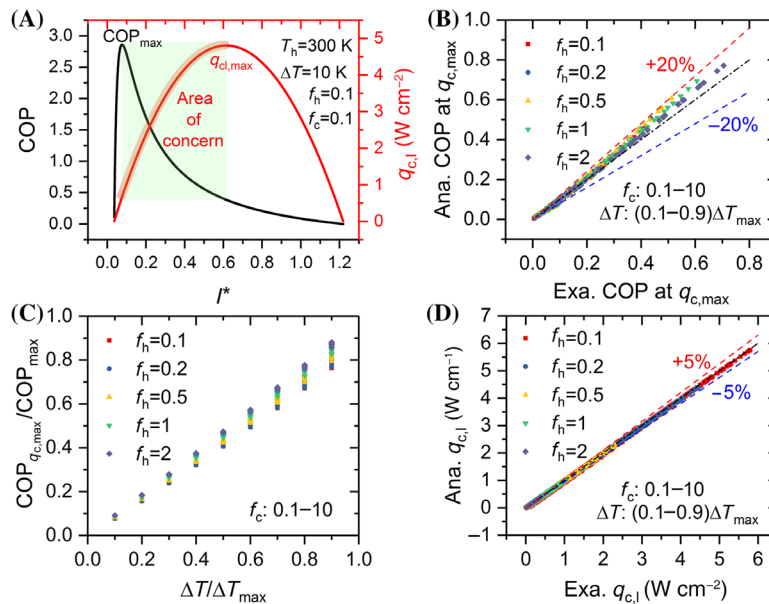


FIGURE 6 (A) Typical curves of COP and cooling power density with the dimensionless electric current under given thermal conditions. (B) Comparison of analytical results of the cooling power maximized COP by Equation (24) with numerical data. (C) Evolutions of the ratio of the cooling power maximized COP to the maximum COP with the dimensionless ΔT . (D) Validation of the approximate formula of q_c (Equation 25) by numerical data

the external heat transfer conditions. Consequently, it is beneficial for a TEC system to work within the intermediate region between the maximum COP point and the maximum cooling power as highlighted in Figure 6A in view of energy utilization efficiency. Interestingly, Lee⁵⁶ suggested to take the electric current be 0.5 times that for maximum cooling power as an optimal operating condition to balance the cooling power and the COP. He further explained that it could provide about half of the maximum cooling power and at least four folds of the cooling power-optimized COP. Nevertheless, it is still a rather rough method only accounting for the TEC itself, without matching the TEC's capability with end-users' practical cooling requirements. Here we offer a comprehensive and direct solution for the maximization of COP under a specified cooling power density at a given temperature difference.

First, it is essential to determine the required electric current for a specified cooling power density, when the external heat transfer conditions and the module parameters are given. Although the exact formula of the cooling power with respect to the electric current (Equation 5) is available, it is too complicated to be analytically solved. From the cooling power density curves (Figure 1B) it can be clearly seen that, in spite of the complicated formula, the curves under different external thermal resistance ratios still evolve virtually as parabolas. Consequently, it is possible to construct a parabolic function as an approximation of the relationship between the cooling power density and the dimensionless electric current within the concerned region in Figure 6A, by utilizing available knowledge of the zero-current point and the maximum cooling power point,

$$q_c = q_{c,0} + 2(q_{c,\max} - q_{c,0}) \frac{I^*}{I_{q_{c,\max},2}^*} - (q_{c,\max} - q_{c,0}) \left(\frac{I^*}{I_{q_{c,\max},2}^*} \right)^2 \quad (25)$$

where

$$q_{c,0} = -\frac{1}{1+f_h+f_c} \frac{\kappa \Delta T}{L^*}$$

and $q_{c,\max}$ is calculated by Equation (10B). To guarantee a good approximation of the cooling power density, the dimensionless electric current at the maximum cooling power must be accurately located. So in Equation (25), the better formula of $I_{q_{c,\max},2}^*$ given by Equation (11) is used instead of the rough one given by Equation (8). As shown in Figure 6D, Equation (25) accurately evaluates the cooling power densities for electric currents between 0 and $I_{q_{c,\max},2}^*$ under various thermal conditions, with

errors in $\pm 5\%$ except for rare cases with extremely low cooling power densities. With the help of Equation (25), we can directly obtain the electric current ratio defined as $r_{I^*} = I^*/I_{q_{c,\max},2}^*$ for a specified cooling power density,

$$r_{I^*} = 1 - \sqrt{\frac{q_{c,\max} - q_c}{q_{c,\max} - q_{c,0}}} \quad (26)$$

with known $I_{q_{c,\max},2}^*$, the required electric current is determined.

Second, we will consider the match of the TEC's capability with a specific cooling requirement in detail. In the following discussion, the peak cooling power will serve as a base point, with which the location of the maximum COP is determined. As discovered in the former section, an optimum L^* is defined for achieving the peak cooling power density under given heat transfer conditions and temperature differences. Now let us specify a target cooling power density, for example, 60% of the peak value. We find that there exists an adoptable region where for any specific L^* , the target cooling power density could be met by providing the proper electric current given by Equation (26), as illustrated in Figure 7. Apparently, the adoptable L^* region will gradually converge toward the $L_{q_{c,\text{peak}}}^*$ as the specified cooling power density approaches the peak value. The COP curve is plotted within the adoptable L^* region, which evidently indicates that there exists a peak COP point for the specified cooling power density. In general, the peak COP point will not coincide with the peak cooling power density point, unless the specified cooling power density is exactly equal to the peak value, thus leaving no room for COP optimization. What needs to be done at last is to

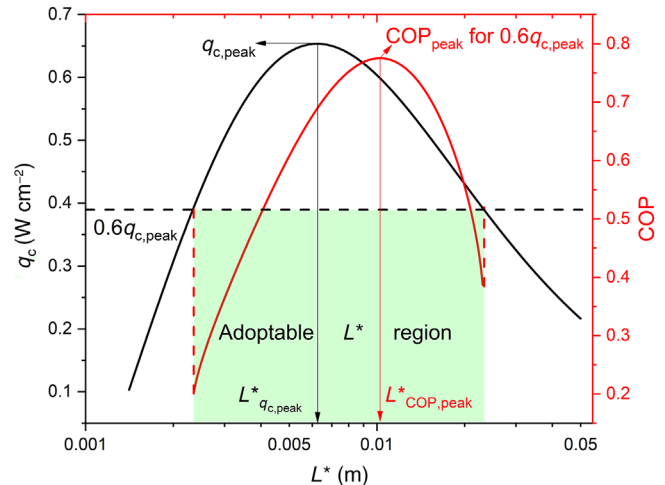


FIGURE 7 Maximization of COP for specified cooling power density under given thermal conditions

find out the value of the peak COP as well as its location on the L^* axis.

Although we have successfully derived a series of approximate formulae with favorable accuracies, we have to admit here that the peak COP under a specified cooling power density and the corresponding L^* are not possible to be analytically solved due to the extremely complex relationships involved. Fortunately, it is possible to construct general explicit correlations for the calculations of COP_{peak} and L_{peak}^* by subtly choosing a base point and relevant influencing parameters. Although the following correlations are not strictly analytical derivations, it will be seen later that they can directly lead to information of the peak COP point just as analytical formulae do. First, L_{peak}^* and COP_{peak} could be represented by two dimensionless parameters,

$$r_{L^*} = \frac{L_{\text{peak}}^*}{L_{q_{\text{c,peak}}}^*}$$

and

$$r_{\text{COP}} = \frac{COP_{\text{peak}}}{COP_{\text{max,i}}}$$

respectively, where $L_{q_{\text{c,peak}}}^*$ is calculated by Equation (22) and $COP_{\text{max,i}}$ is the maximum COP under ideal thermal conditions given by Equation (1C). Second, r_{L^*} and r_{COP} are supposed to be dominated by the same set of dimensionless influencing parameters,

$$r_h = \frac{h_h}{h_c}$$

$$r_{q_c} = \frac{q_c}{q_{\text{c,peak}}}$$

and

$$r_{\Delta T} = \frac{\Delta T}{\Delta T_{\text{max,i}}}$$

where $q_{\text{c,peak}}$ is calculated by substituting Equation (22) into Equation (21), and $\Delta T_{\text{max,i}}$ is the maximum ΔT under ideal thermal conditions given by Equation (1B). Then, a concise and reasonable formula is constructed for each undetermined parameter in the following way. Remember that the COP_{peak} point will coincide with the $q_{\text{c,peak}}$ point on the L^* axis if the specified q_c is equal to $q_{\text{c,peak}}$, that is, $r_{L^*} = 1$ when $r_{q_c} = 1$, regardless of r_h and $r_{\Delta T}$. This base point inspires us to adopt the following formula to the general correlation of r_{L^*} ,

$$r_{L^*} = r_{q_c}^f(r_h, r_{q_c}, r_{\Delta T}) \quad (27)$$

Equation (27) strictly obeys the above constraint condition. Now let us go back to Figure 6C, which implies that r_{COP} will be close to 1 (but will never reach 1 under practical heat transfer conditions) as $r_{\Delta T}$ approaches 1, in a similar manner as the evolution of r_{L^*} with r_{q_c} . Hence, we will use the following basic form in the construction of the correlation of r_{COP} ,

$$r_{\text{COP}} = r_{\Delta T}^g(r_h, r_{q_c}, r_{\Delta T}) \quad (28)$$

which follows the constraint condition in a less strict way as Equation 27, since the constraint condition here only describes a qualitative trend. This could be a good choice, as will be shown later. $f(r_h, r_{q_c}, r_{\Delta T})$ and $g(r_h, r_{q_c}, r_{\Delta T})$ in Equations (27) and (28) could be expressed as polynomials of the relevant parameters, whose orders can be adjusted to achieve the required accuracy. We have found that such a second-order polynomial as

$$\begin{aligned} & a_0 + a_1 r_h + a_2 r_h^2 + b_1 r_{q_c} + b_2 r_{q_c}^2 + c_1 r_{\Delta T} + c_2 r_{\Delta T}^2 + d_1 r_h r_{q_c} \\ & + d_2 r_{q_c} r_{\Delta T} + d_3 r_h r_{\Delta T} + d_4 r_h r_{q_c} r_{\Delta T} \end{aligned}$$

is enough to evaluate r_{L^*} and r_{COP} with accuracies of $\pm 15\%$ over a broad range of heat transfer conditions and cooling requirements, as shown in Figure 8. In very few cases with very high $r_{\Delta T}$ (above 0.8), r_{COP} is evaluated with errors between 15% and 25% (see Figure 8B).

The coefficients in $f(r_h, r_{q_c}, r_{\Delta T})$ and $g(r_h, r_{q_c}, r_{\Delta T})$ are obtained by using the least square method over a large amount of numerically induced results within the selected operating range and listed in Table 1. Although the coefficients shown here are obtained using the state-of-the-art TE materials with properties of $S = 200 \mu\text{V K}^{-1}$, $\sigma = 1\text{e}5 \text{ S m}^{-1}$, $\kappa = 1.5 \text{ W m}^{-1} \text{ K}^{-1}$ and $Z = 0.0026$, it is good to see that they can make reasonable predictions of r_{L^*} and r_{COP} even when the figure of merit of the thermoelectric material is increased by 200% ($Z = 0.008$, see Figure S5). Consequently, Equations (27) and (28) pave a direct way for solving the COP maximization problem under given heat transfer conditions and cooling requirements.

3.3 | Optimization of thermal resistance allocation

From the above discussion, we realize that the peak cooling power is unsuitable for direct use in most

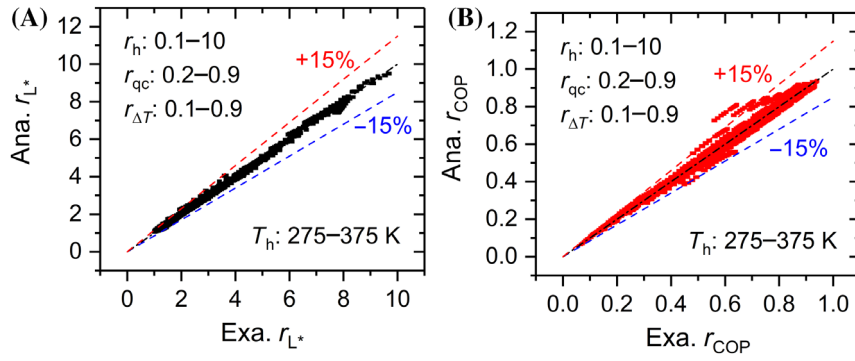


FIGURE 8 Validations of correlations of (A) r_{L^*} (Equation 27) and (B) r_{COP} (Equation 28) by numerical data

TABLE 1 Coefficients in expressions for f and g in Equations (27) and (28)

	a_0	a_1	a_2	b_1	b_2	c_1	c_2	d_1	d_2	d_3	d_4
f	-0.4895	-0.02066	1.5139E-3	1.1522	-1.0243	-1.7875	1.1532	-0.1733	-1.4794	3.484E-4	0.1431
g	0.3274	0.03981	-2.6245E-3	0.1610	0.4659	-1.7887	1.9471	0.01531	1.4451	-0.03956	0.1399

situations. However, it is still beneficial to raise it up as high as possible under specific heat transfer constraints, because a higher peak cooling power will leave more room for the optimization of COP under given cooling requirements, as illustrated in Figure 7. In this section, we will look into the thermal resistance allocation, which is critical for the peak cooling power of a TEC system. Allocating the available heat transfer area or heat transfer coefficient at the hot and cold sides of the module is meaningful as the hot and cold side heat exchangers are intercorrelated such that a rise in one heat exchanger's capability must be at the cost of a reduction in that of the other. Such kind of coupling between the heat exchangers often occurs in practical applications as a total constraint of cost, weight, or space.

In previous studies, the constraint is often simplified as a total heat transfer area shared by the two heat exchangers,^{28–31} which is convenient to handle and obtain quantitative results. In this work, we will follow this treatment in a slightly different way. Since we represented the heat transfer capability as the heat transfer coefficient based on the substrate area of the TEC module in former derivations, we will describe the constraint as a finite total heat transfer coefficient shared by the two heat exchangers, that is,

$$h_h + h_c = h_{tot} \quad (29)$$

which is equivalent to the total heat transfer area constraint with a uniform heat transfer coefficient on it. Now the key is to find the optimum allocation ratio (r_h)

between h_h and h_c to achieve the peak cooling power. We have already deduced the peak cooling power density under given heat transfer conditions as Equation (21). Substituting Equation (22) into Equation (21), we can get a more detailed form,

$$q_{c,peak} = \frac{r_h h_{tot}}{1 + r_h} \frac{0.5 Z T_c^2 \left[1 + \frac{Z T_c}{\phi(1 + 2\psi)^2} \right] - (1 + 2\psi)\Delta T}{(\phi + 2)[1 + (1 + r_h)\psi]} \quad (30)$$

where $\psi = (1 + Z T_c)/\phi$. ϕ is also dependent on r_h , so we can obtain the optimum r_h by letting $dq_{c,peak}/dr_h = 0$ under a constant h_{tot} . Again, this derivative is too complicated to be analytically solved. Whereas, the involved parameters are limited when we look carefully into Equation (30), which could be well represented by two dimensionless parameters $r_{\Delta T}$ and $Z T_h$. We choose $Z T_h$ instead of $Z T_c$ for convenient use since T_h , the heat sink temperature, is directly known in practical situations. First, let us glance over the evolutions of the optimal r_h with $r_{\Delta T}$ and $Z T_h$ as provided in Figure 9A. The optimal r_h is only weakly affected by $Z T_h$, while it experiences a sharp rise when $r_{\Delta T}$ gets close to 1. This is reasonable since the achievable ΔT_{max} in real situations is dominated by the hot side heat transfer condition as shown in Equation (13), so more heat transfer coefficient should be allocated at the hot side under higher $r_{\Delta T}$ conditions. When $r_{\Delta T}$ infinitely approaches 1, no positive cooling power will be produced even with all of the finite heat transfer

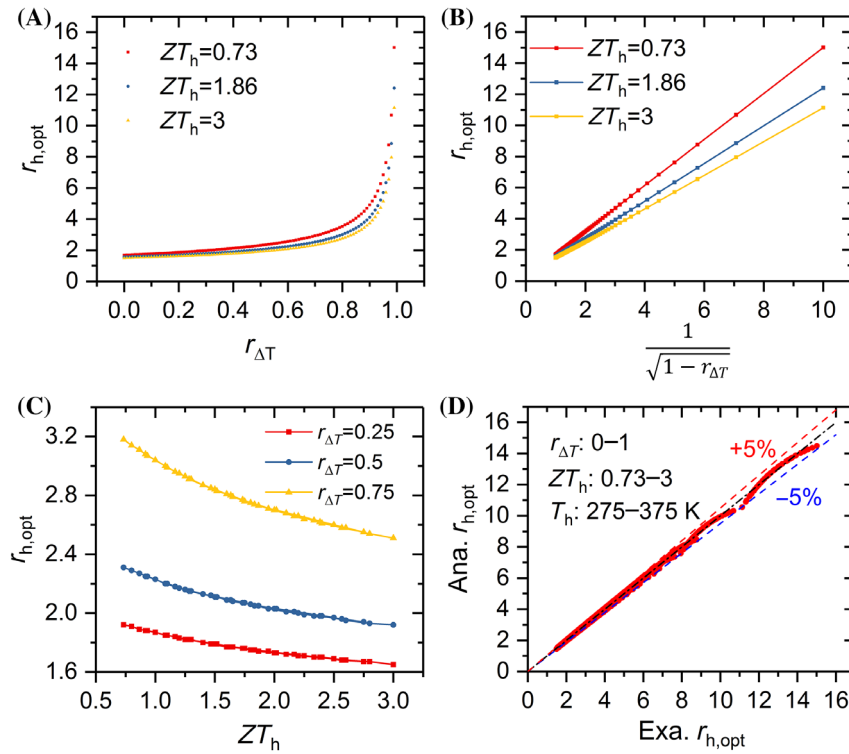


FIGURE 9 Variations of the optimal r_h for peak cooling power density with (A) $r_{\Delta T}$, (B) $1/\sqrt{1-r_{\Delta T}}$ and (C) ZT_h ; (D) validation of Equation (31) on the evaluation of the optimal r_h

coefficient being put at the hot side, which means that the optimum r_h will go to infinity. Considering this singular point, it is plausible to postulate a term in the final correlation accounting for $r_{\Delta T}$ as $1/(1-r_{\Delta T})$. Based on this postulation, we further find that the optimum r_h is almost linear with $1/\sqrt{1-r_{\Delta T}}$ under a constant ZT_h , as shown in Figure 9B. Moreover, the curves with varying ZT_h under given $r_{\Delta T}$ in Figure 9C indicate that the optimum r_h weakly decreases with ZT_h also in an approximately linear manner. The varying ZT_h in this figure is produced by changing T_h in a range of 275–375 K and Z in a range of 0.0026–0.008, with an upper limit of three times of the state-of-the-art Z value. Consequently, the discussion here and the following correlation of the optimal r_h are valid even with thermoelectric materials' performance being significantly increased in the future. Finally, we construct a correlation as

$$r_{h,opt} = a_0 + a_1 \frac{1}{\sqrt{1-r_{\Delta T}}} + b_1 ZT_h + c_1 \frac{ZT_h}{\sqrt{1-r_{\Delta T}}} \quad (31)$$

where the coefficients, a_0 , a_1 , b_1 , and c_1 are obtained using the least square method and listed in Table 2. The results given by Equation (31) are compared with numerical data in Figure 9D over a broad range, demonstrating an excellent accuracy (errors within $\pm 5\%$). Some degree of nonlinearity exists between $r_{h,opt}$ and ZT_h especially

TABLE 2 Coefficients in Equation (31)

a_0	a_1	b_1	c_1
0.1818	1.5574	0.08528	−0.1822

for higher $r_{h,opt}$ levels (see Figure 9C,D), so a correlation of a higher order of ZT_h shall lead to an even better accuracy. Nevertheless, we will stop at Equation (31) since it is simple and effective. Very high allocation ratios are rarely met in practical applications, so there is no need to pay more attention.

3.4 | A general design procedure for TEC systems

In the former sections, we have comprehensively discussed the three important optimizing case regarding a TEC system and proposed a set of explicit formulae to enable straightforward evaluations of involved parameters. Now it is time to establish a general object-oriented design procedure for TEC systems. We present it schematically in Figure 10 and explain it in detail as follows. Briefly, this design procedure can be used for determinations of such parameters as the heat transfer conditions (h_h and h_c) outside the TEC module, the module internal

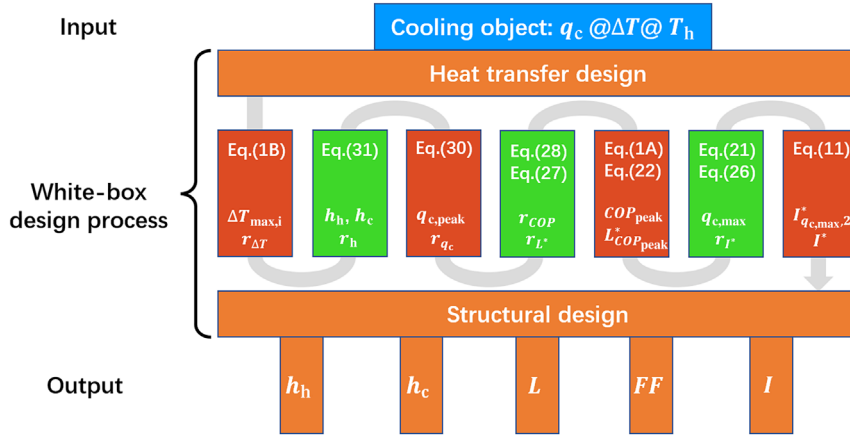


FIGURE 10 A general object-oriented design procedure for TEC systems

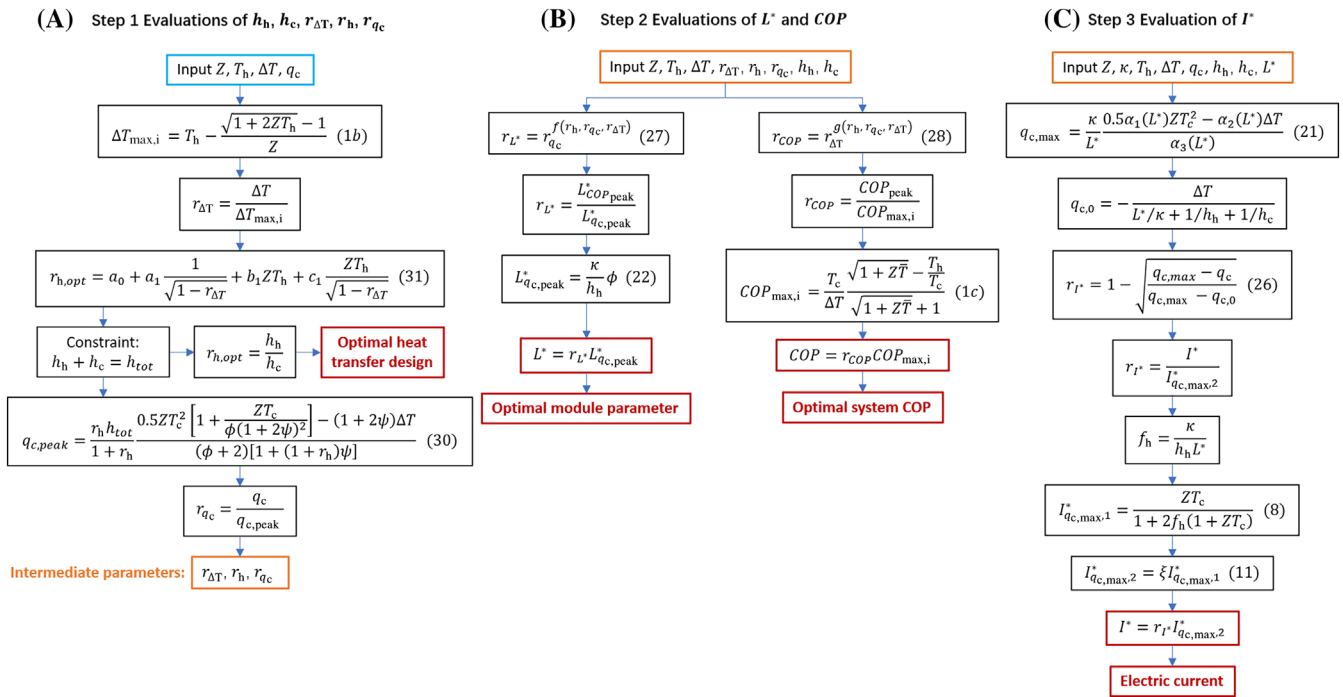


FIGURE 11 A detailed flowchart of the design procedure

dimensions including the leg length (L) and the fill factor (FF), and the required electric current (I) to meet the cooling requirement of an end-user with the peak COP. A specified cooling power density (q_c) represents the specific cooling requirement over a temperature difference (ΔT), under a given heat sink temperature (T_h). To make the abstract design procedure more intuitive, we provide a detailed flowchart in Figure 11 and carry out a concrete design case along with each design step (see SI: Design procedure for TEC systems for detailed information).

The whole design procedure is divided into three steps, as shown in Figure 11. In the first step, three dimensionless parameters, that is, $r_{\Delta T}$, r_h , and r_{q_c} as well

as the heat transfer coefficients (h_h and h_c) are determined. With specified material properties, ΔT and T_h , the value of $r_{\Delta T}$ is determined with Equation (1B) and its definition formula, $r_{\Delta T} = \Delta T / \Delta T_{\max,i}$. The heat transfer coefficients are allocated with the optimal ratio ($r_h = h_h / h_c$) according to Equation (31) together with the total available heat transfer resource under certain constraints. Then the peak cooling power density ($q_{c,peak}$) is calculated by Equation (30), which sets the value to r_{q_c} together with the specified q_c .

The second step focuses on the evaluations of L^* and COP. Now that we have determined the values of r_h , r_{q_c} , and $r_{\Delta T}$, we can move forward to evaluate the local peak

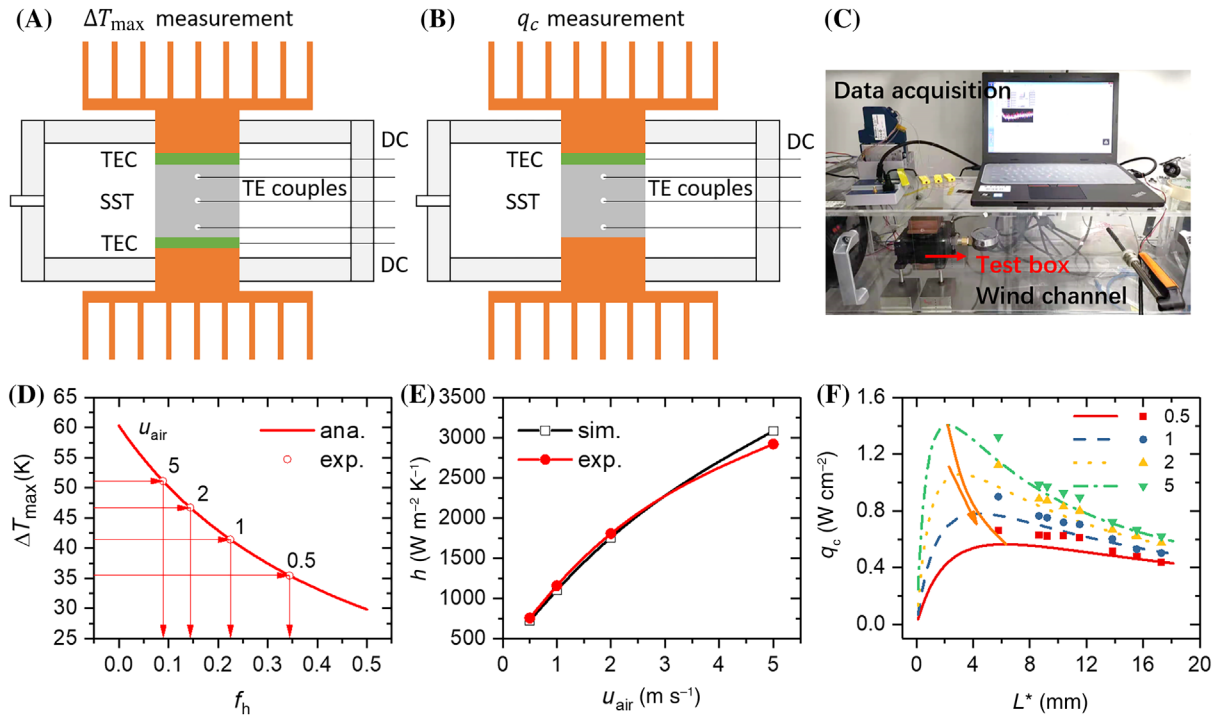


FIGURE 12 (A) test box arrangement for $\Delta T_{\max}$ measurement; (B) test box arrangement for q_c measurement; (C) the whole test system; (D) evaluations of the dimensionless thermal resistance parameters of the heat exchanger according to the measured $\Delta T_{\max}$ under different air velocities; (E) evaluations of the heat transfer coefficients of the heat exchanger under different air velocities; (F) evolutions of the maximum cooling power density versus L^* under different air velocities, curves for analytical results and scatters for measured data

COP ratio (r_{COP}) under given cooling requirements and the corresponding L^* ratio (r_{L^*}) according to Equations (27) and (28). The denominators of the above two ratios, $\text{COP}_{\max,i}$ and $L^*_{q_{c,\text{peak}}}$, are evaluated by Equations (1A) and (22), respectively, so the numerators, COP_{peak} and $L^*_{\text{COP}_{\text{peak}}}$ are obtained.

The dimensionless electric current is evaluated in the third step. Taking $L^*_{\text{COP}_{\text{peak}}}$ into Equation (21), we get the maximum cooling power density $q_{c,\text{max}}$ under the selected L^* , and further get the dimensionless electric current ratio r_{I^*} by Equation (26). The denominator of r_{I^*} , $I^*_{q_{c,\text{max},2}}$, is calculated by Equation (11), so the required dimensionless electric current I^* is obtained.

h_h , h_c , L^* , and I^* are the outcomings of the above design procedure, which leaves some free room for the determinations of L and FF in terms of other factors such as the structural feasibility. So, the final system design is not unique, but could be chosen from a basket of available concrete designs with no essential difference in the cooling performance. With given L and FF , the leg thermal resistance $R_{t,l}$ is obtained. Divided by $SR_{t,l}$, I^* finally assigns a definite value to the input electric current I . In case of time-varying cooling requirements, an adoptable L^* range should be found at first to cope with the worst scenario with the highest cooling power density or temperature difference under determined heat transfer

conditions. Then the optimal L^* for peak COP under the most common cooling requirement if any, or that for the maximum time- and cooling power-weighted COP during the whole working time could be extracted as the final choice from the adoptable range.

At last, we will experimentally go through some critical steps of the design procedure, including evaluations of thermal resistances of the heat exchangers and evolutions of the maximum cooling power density with L^* under given thermal environments. When we design or choose heat exchangers for a TEC system, we need to determine their heat transfer coefficients under specific thermal circumstances. Here we will do this work in two ways. The first way is using numerical simulations (with the commercial software FLUENT as the simulation tool), while the second way is based on a method derived from our analytical framework. From the formula of $\Delta T_{\max}$ under real heat transfer conditions (Equation 13), the hot side thermal resistance ratio can be deduced as

$$f_h = \frac{ZT_c^2 - 2\Delta T_{\max}}{4\Delta T_{\max}(1 + ZT_c)} \quad (32)$$

Once $\Delta T_{\max}$ is measured under given heat sink temperature, we can evaluate f_h by Equation (32). With the known thermal resistance of the TEC module, the heat

transfer coefficient based on the substrate area can be obtained. To perform the $\Delta T_{\max}$ and the following cooling power measurements, we have constructed a test box as shown in Figure 12A. During the $\Delta T_{\max}$ measurements, two TEC modules with the same material properties and dimensional parameters are mounted in the vacuum chamber, and two heat exchangers with the same parameters are mounted symmetrically outside the vacuum chamber, with their bases penetrating into the chamber and connecting to the two TEC modules. Due to the symmetric configuration of the TEC modules and heat exchangers, an adiabatic condition is created at the cold side of the TEC modules so that the $\Delta T_{\max}$ can be obtained. During the cooling power measurements, only one TEC module is placed in the vacuum chamber (see Figure 12B). A stainless steel-heat flow meter (thermal conductivity: $\sim 15 \text{ W m}^{-1} \text{ K}^{-1}$) is inserted between the TEC module and the cold side heat exchanger, and the cooling power is calculated by the Fourier thermal conduction law with the readings of the thermocouples inside the heat flow meter. During tests, the test box is placed in a wind channel in which the air temperature and velocity can be precisely controlled (see Figure 12C). Since the hot and cold sides of the TEC modules are under the same convection condition (meaning $\Delta T = 0$) in the present arrangement, it can be seen from Equations (14 and (15 that the maximum COP will be infinite with an electric current of zero, so it will not be measured in the following tests.

A copper-made flat-fin heat exchanger is designed for the TEC system, whose dimensional parameters can be found in Table S4. Eight home-made commercial Bi_2Te_3 -based TEC modules with different L^* values ranging from 5 to 20 mm are prepared by changing their leg lengths and fill factors (see Table S5 for the modules' dimensional parameters). The module with a L^* value of 5.767 mm is used to perform the $\Delta T_{\max}$ measurements under different air velocities while stable ambient temperature of 295.15 K, from which the dimensionless thermal resistance ratios of the heat exchanger are estimated using Equation (32) (Figure 12D). The heat transfer coefficients of the heat exchanger are further evaluated and compared with the numerically obtained values, as shown in Figure 12E. Good agreement is reached between the test and numerical results by setting the average values of the physical properties as $S = 200 \mu\text{V K}^{-1}$, $\sigma = 0.82 \text{ eS m}^{-1}$, $\kappa = 1.5 \text{ W m}^{-1} \text{ K}^{-1}$, which gives $Z = 0.0022$ in Equation (32). So we will use these values in the evaluations of the maximum cooling power densities. Figure 12F presents evolutions of the maximum cooling power density versus L^* under different air velocities, with both analytical curves calculated by Equation (21) and test results. Reasonable agreement is reached between the analytical and measured results. The migration of the peak cooling power density point toward

higher L^* values with decreasing air velocities is clearly seen, confirming a strong dependence of the design of the TEC module on the external thermal circumstance.

4 | CONCLUSION

Thermal resistances of the module and those at the hot and the cold sides are critical to the design of TECs for optimal cooling performance. We have developed an analytical framework in this work consisting of a set of explicit formulae. As a result, the performances of a TEC system in practical thermal circumstances can be evaluated in a transparent manner consistent with physical intuitions. Based on the framework, we have further discussed the areas of most concern in the design of TEC systems, including the maximization of the cooling power under given heat transfer conditions, the optimization of COP under given cooling power and temperature difference, and the optimal allocation of heat transfer capability at the two sides of the module for maximum cooling power. With the construction of a group of explicit formulae and correlations concerning the vital topics, we have liberated the TEC system design from the obscure and time-consuming numerical solution method by a White-Box and straightforward procedure. Our work is hopeful to provide a standard path for estimating and designing TEC systems and releasing the full potential of advanced thermoelectric materials in the cooling field in broad demand in the modern world.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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NOMENCLATURE

S	Seebeck coefficient
Z	figure of merit
T	temperature

L	leg length
A	leg cross-section area
A_s	substrate area
FF	fill factor
L^*	engineering leg length
N	number of TE couples
I	electric current
I^*	dimensionless electric current
R	leg electrical resistance
$R_{t,l}$	leg thermal resistance
h	heat transfer coefficient
f	dimensionless thermal resistance parameter
ΔT	external temperature difference
$\Delta T'$	internal temperature difference
Q_c	cooling power
P	electric power
COP	coefficient of performance
q_{cl}	leg based cooling power density
r_h	heat transfer coefficient ratio
$r_{R_{t,m},q_{c,peak}}$	optimal thermal resistance ratio
r_{L^*}	engineering leg length ratio
r_{COP}	coefficient of performance ratio
r_{q_c}	cooling power density ratio

GREEK SYMBOLS

σ	electrical conductivity
κ	thermal conductivity

SUBSCRIPTS

h	heat sink
c	cooling target
h'	hot side of the module
c'	cold side of the module
i	ideal
l	leg
m	module
max	maximum point
opt	optimal
peak	peak point
s	substrate
tot	total

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SUPPORTING INFORMATION

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